

Project Screenshots

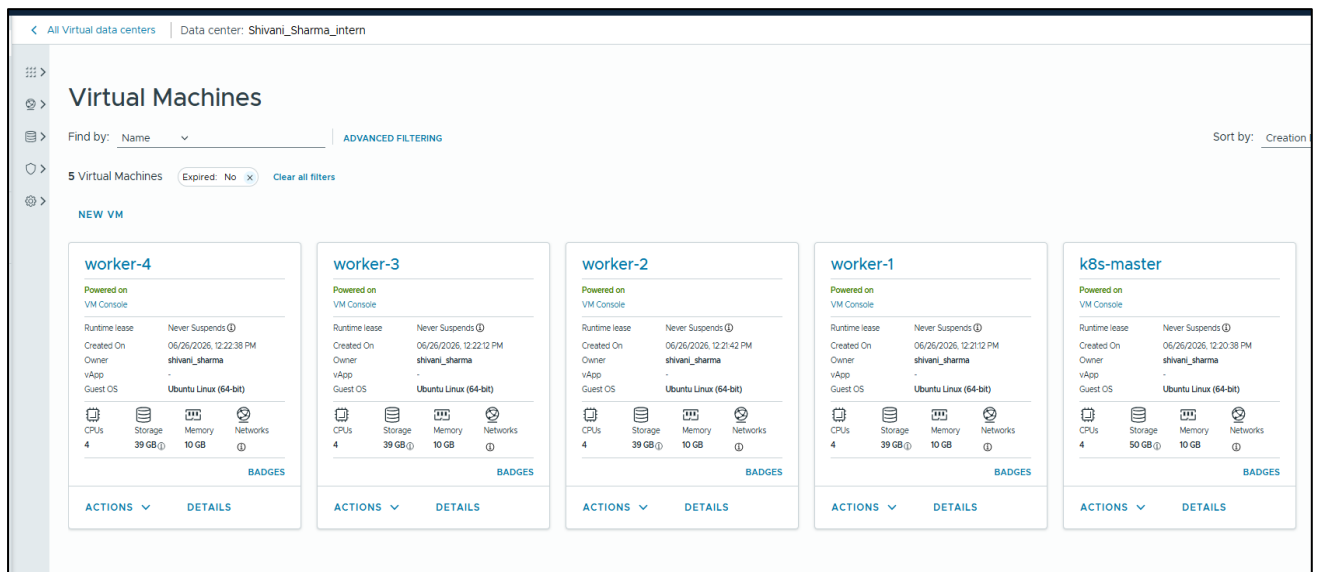


Figure: VMs Configuration.

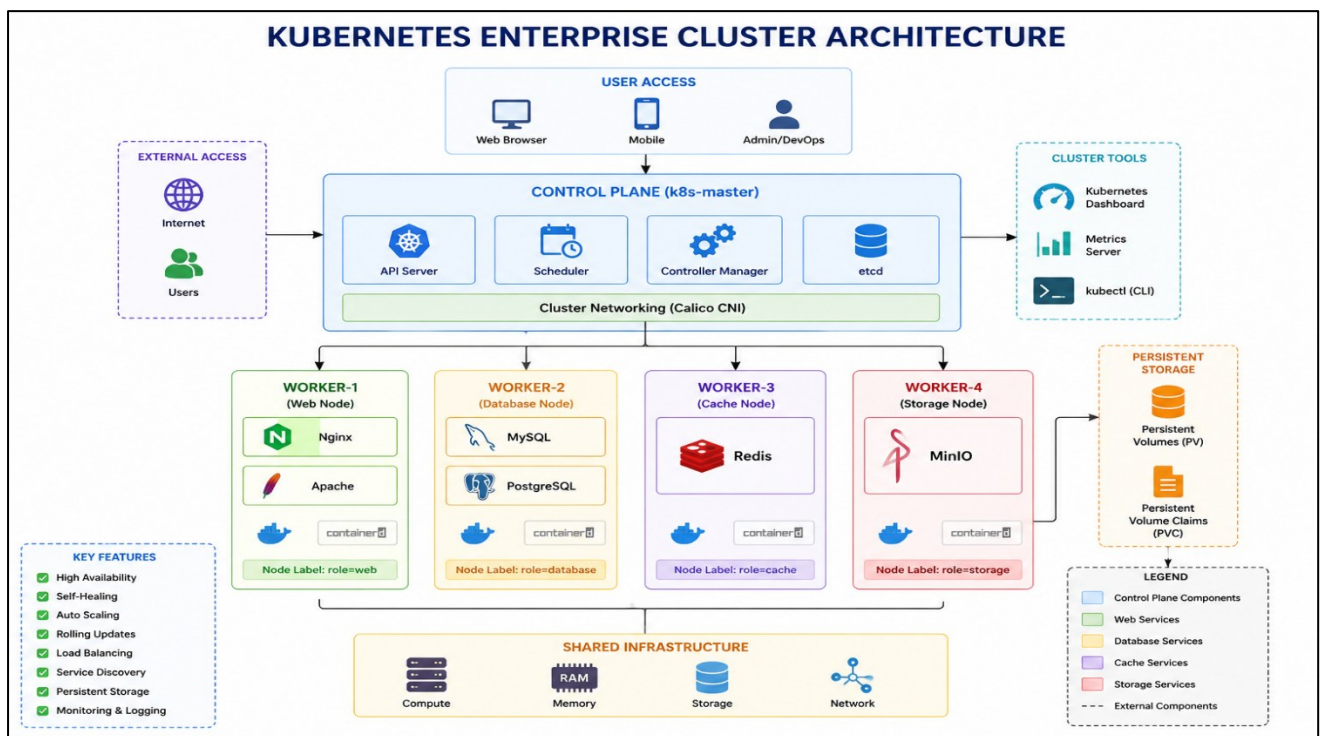


Figure: Project Architecture.

```
root@k8s-master:~# swapoff -a
sed -i '/swap/d' /etc/fstab
root@k8s-master:~# free -h
```

	total	used	free	shared	buff/cache	available
Mem:	9.7Gi	376Mi	4.8Gi	1.0Mi	4.5Gi	9.0Gi
Swap:	0B	0B	0B			

```
root@k8s-master:~#
```

Figure: Disable swap

```

root@k8s-master:~# cat <<EOF | tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF
root@k8s-master:~# modprobe overlay
modprobe br_netfilter
root@k8s-master:~#

```

Figure: Enable Kernel Modules

```

root@k8s-master:~# cat <<EOF | tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables=1
net.bridge.bridge-nf-call-ip6tables=1
net.ipv4.ip_forward=1
EOF
root@k8s-master:~# sysctl --system
* Applying /etc/sysctl.d/10-console-messages.conf ...
kernel.printk = 4 4 1 7
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
kernel.kptr_restrict = 1
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
kernel.sysrq = 176
* Applying /etc/sysctl.d/10-network-security.conf ...
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
* Applying /etc/sysctl.d/10-ptrace.conf ...
kernel.yama.ptrace_scope = 1
* Applying /etc/sysctl.d/10-zero-page.conf ...
vm.mmap_min_addr = 65536
* Applying /usr/lib/sysctl.d/50-default.conf ...
kernel.core_uses_pid = 1

```

Figure: Configure Network Parameters

```

root@k8s-master:~# mkdir -p /etc/containerd
root@k8s-master:~# containerd config default > /etc/containerd/config.toml
root@k8s-master:~# nano /etc/containerd/config.toml
root@k8s-master:~# systemctl restart containerd
systemctl enable containerd
root@k8s-master:~# systemctl status containerd
● containerd.service - containerd container runtime
   Loaded: loaded (/lib/systemd/system/containerd.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2026-06-26 07:23:52 UTC; 11s ago
     Docs: https://containerd.io
   Main PID: 54631 (containerd)
    Tasks: 9
   Memory: 14.5M
     CPU: 72ms
   CGroup: /system.slice/containerd.service
           └─54631 /usr/bin/containerd

```

Figure: Install Container Runtime

```

root@k8s-master:~# mkdir -p /etc/apt/keyrings
root@k8s-master:~# curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.30/deb/Release.key | gpg --dearmor -o /etc/apt/keyring
root@k8s-master:~# echo "deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:/stable:/v
.d/kubernetes.list
root@k8s-master:~# apt update
Hit:2 http://archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://archive.ubuntu.com/ubuntu jammy-updates InRelease
Get:1 https://prod-cdn.packages.k8s.io/repositories/ipv:/kubernetes:/core:/stable:/v1.30/deb InRelease [1,192 B]
Get:4 https://prod-cdn.packages.k8s.io/repositories/ipv:/kubernetes:/core:/stable:/v1.30/deb Packages [20.3 kB]
Hit:5 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:6 http://archive.ubuntu.com/ubuntu jammy-security InRelease
Fetched 21.5 kB in 2s (12.5 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
root@k8s-master:~# apt install kubelet kubeadm kubectl -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libfwupdplugin5 libgcb-1.0-0 libgusb2 libmbim-glib4 libmbim-proxy libmm-glib0 libqmi-glib5 libqmi-proxy libst
  tcl tcl8.6 usb-modeswitch usb-modeswitch-data
Use 'apt autoremove' to remove them.
The following additional packages will be installed:
  conntrack cri-tools kubernetes-cni
The following NEW packages will be installed:
  conntrack cri-tools kubeadm kubectl kubelet kubernetes-cni
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 93.8 MB of archives.
After this operation, 343 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 conntrack amd64 1:1.4.6-2build2 [33.5 kB]
Get:2 https://prod-cdn.packages.k8s.io/repositories/ipv:/kubernetes:/core:/stable:/v1.30/deb cri-tools 1.30.1-1.1 [21.3 M
Get:3 https://prod-cdn.packages.k8s.io/repositories/ipv:/kubernetes:/core:/stable:/v1.30/deb kubeadm 1.30.14-1.1 [10.5 M
Get:4 https://prod-cdn.packages.k8s.io/repositories/ipv:/kubernetes:/core:/stable:/v1.30/deb kubectl 1.30.14-1.1 [10.9 M

```

Figure: Install Kubernetes Packages

```

[kubelet-finalize] Updating "/etc/kubernetes/kubelet.conf" to point to a rotatab
le kubelet client certificate and key
[addons] Applied essential addon: CoreDNS
[addons] Applied essential addon: kube-proxy

Your Kubernetes control-plane has initialized successfully!

To start using your cluster, you need to run the following as a regular user:

  mkdir -p $HOME/.kube
  sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
  sudo chown $(id -u):$(id -g) $HOME/.kube/config

Alternatively, if you are the root user, you can run:

  export KUBECONFIG=/etc/kubernetes/admin.conf

You should now deploy a pod network to the cluster.
Run "kubectl apply -f [podnetwork].yaml" with one of the options listed at:
  https://kubernetes.io/docs/concepts/cluster-administration/addons/

Then you can join any number of worker nodes by running the following on each as
root:

kubeadm join 192.168.0.2:6443 --token ms6gt0.d41n33gxhc5u4364 \
  --discovery-token-ca-cert-hash sha256:2b546d251c51b9463c5895f6e6c87bdae9
39319e6cc6d0a1679739789b362e35

```

Figure: Initialize Control Plane

```

root@worker-1:~# kubeadm join 192.168.0.2:6443 \
  --token ms6gt0.d4ln33gxhc5u4364 \
  --discovery-token-ca-cert-hash sha256:2b546d251c51b9463c5895f6e6c87bd9e939319e6cc6d0a1679739789b362e35
[preflight] Running pre-flight checks
[preflight] Reading configuration from the cluster...
[preflight] FYI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubelet-start] Writing kubelet configuration to file "/var/lib/kubelet/config.yaml"
[kubelet-start] Writing kubelet environment file with flags to file "/var/lib/kubelet/kubeadm-flags.env"
[kubelet-start] Starting the kubelet
[kubelet-check] Waiting for a healthy kubelet at http://127.0.0.1:10248/healthz. This can take up to 4m0s
[kubelet-check] The kubelet is healthy after 501.823731ms
[kubelet-start] Waiting for the kubelet to perform the TLS Bootstrap

This node has joined the cluster:
* Certificate signing request was sent to apiservert and a response was received.
* The Kubelet was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.

root@worker-1:~#

```

```

root@k8s-master:~# kubectl get nodes

```

NAME	STATUS	ROLES	AGE	VERSION
k8s-master	Ready	control-plane	8m53s	v1.30.14
worker-1	Ready	<none>	33s	v1.30.14
worker-2	Ready	<none>	102s	v1.30.14
worker-3	Ready	<none>	2m43s	v1.30.14
worker-4	Ready	<none>	6m3s	v1.30.14

```

root@k8s-master:~#

```

```

root@k8s-master:~# kubectl get pods -A

```

NAMESPACE	NAME	READY	STATUS	RESTARTS	AGE
kube-system	calico-kube-controllers-5b64744bcd-65dd7	1/1	Running	0	20m
kube-system	calico-node-4cp69	1/1	Running	0	13m
kube-system	calico-node-54stq	1/1	Running	0	12m
kube-system	calico-node-7kfjq	1/1	Running	0	20m
kube-system	calico-node-md7r5	1/1	Running	0	14m
kube-system	calico-node-x9hz7	1/1	Running	0	18m
kube-system	coredns-55cb58b774-kcsn8	1/1	Running	0	20m
kube-system	coredns-55cb58b774-s6tv7	1/1	Running	0	20m
kube-system	etcd-k8s-master	1/1	Running	0	20m
kube-system	kube-apiserver-k8s-master	1/1	Running	0	20m
kube-system	kube-controller-manager-k8s-master	1/1	Running	0	20m
kube-system	kube-proxy-2nj2w	1/1	Running	0	12m
kube-system	kube-proxy-c4k57	1/1	Running	0	13m
kube-system	kube-proxy-hnnmb	1/1	Running	0	20m
kube-system	kube-proxy-pmk9g	1/1	Running	0	14m
kube-system	kube-proxy-wvtmq	1/1	Running	0	18m
kube-system	kube-scheduler-k8s-master	1/1	Running	0	20m

```

root@k8s-master:~#

```

Figure: Join worker nodes

```

root@k8s-master:~# kubectl label node worker-1 role=web
kubectl label node worker-2 role=database
kubectl label node worker-3 role=storage
kubectl label node worker-4 role=apps
node/worker-1 labeled
node/worker-2 labeled
node/worker-3 labeled
node/worker-4 labeled
root@k8s-master:~# kubectl get nodes --show-labels

```

NAME	STATUS	ROLES	AGE	VERSION	LABELS
k8s-master	Ready	control-plane	22m	v1.30.14	beta.kubernetes.io/arch=amd64, beta.kubernetes.io/os=linux, kubernetes.io/cluster=kubernetes, kubernetes.io/role=master, node-role.kubernetes.io/control-plane=
worker-1	Ready	<none>	14m	v1.30.14	beta.kubernetes.io/arch=amd64, beta.kubernetes.io/os=linux, role=web
worker-2	Ready	<none>	15m	v1.30.14	beta.kubernetes.io/arch=amd64, beta.kubernetes.io/os=linux, role=database
worker-3	Ready	<none>	16m	v1.30.14	beta.kubernetes.io/arch=amd64, beta.kubernetes.io/os=linux, role=storage
worker-4	Ready	<none>	19m	v1.30.14	beta.kubernetes.io/arch=amd64, beta.kubernetes.io/os=linux, role=apps

```

root@k8s-master:~#

```

Figure: Node Labeling

```

root@k8s-master:~# kubectl get ns
NAME                STATUS    AGE
cache               Active    8s
database            Active    8s
default             Active    44m
kube-node-lease     Active    44m
kube-public         Active    44m
kube-system         Active    44m
metallb-system      Active    16m
storage             Active    8s
web                 Active    8s
root@k8s-master:~#

```

Figure: Namespaces

```

root@k8s-master:~# nano nginx.yaml
root@k8s-master:~# kubectl apply -f nginx.yaml
deployment.apps/nginx created
service/nginx-service created

```

```

root@k8s-master:~# kubectl get pods -n web -o wide
NAME                READY    STATUS    RESTARTS   AGE    IP             NODE        NOMINATED NODE    READINESS GATES
nginx-66ccff7bd8-fgdk 1/1      Running   0           31s    192.168.226.66 worker-1     <none>             <none>
nginx-66ccff7bd8-lb6dw 1/1      Running   0           31s    192.168.226.65 worker-1     <none>             <none>
root@k8s-master:~# kubectl get svc -n web
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
nginx-service       LoadBalancer 10.106.79.71   192.168.0.240 80:32643/TCP     84s

```

Figure: Nginx Deployment.

```

root@k8s-master:~# kubectl get pods -n web -o wide
NAME                READY    STATUS    RESTARTS   AGE    IP             NODE        NOMINATED NODE
apache-8875fdb8c8-btxvc 1/1      Running   0           51s    192.168.226.68 worker-1     <none>
apache-8875fdb8c8-cfnkf 1/1      Running   0           51s    192.168.226.67 worker-1     <none>
nginx-66ccff7bd8-fgdk 1/1      Running   0           9m4s   192.168.226.66 worker-1     <none>
nginx-66ccff7bd8-lb6dw 1/1      Running   0           9m4s   192.168.226.65 worker-1     <none>
root@k8s-master:~# kubectl get svc -n web
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
apache-service      LoadBalancer 10.99.101.178 192.168.0.241 80:30795/TCP     2m38s
nginx-service       LoadBalancer 10.106.79.71  192.168.0.240 80:32643/TCP     10m
root@k8s-master:~# curl 192.168.0.241
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">
<html>
<head>
<title>It works! Apache httpd</title>
</head>
<body>
<p>It works!</p>
</body>
</html>

```

Figure: Apache Deployment.

```

root@k8s-master:~# kubectl get pods -n database -o wide
NAME                READY    STATUS    RESTARTS   AGE    IP             NODE        NOMINATED NODE
mysql-676f465746-97jj7 1/1      Running   0           3m     192.168.133.193 worker-2     <none>
postgres-757885f655-8hv9m 1/1      Running   0           2m7s   192.168.133.194 worker-2     <none>
root@k8s-master:~#

```

Figure: MySQL Deployment.


```

root@k8s-master:~# kubectl get pods -A -o wide
NAMESPACE   NAME                                     READY   STATUS    RESTARTS   AGE   IP              NODE           NOMINATED NODE   READINESS GATES
S GATES
cache        redis-7b8446847d-hhtx5                 1/1     Running   0           11s   192.168.38.69   worker-4       <none>           <none>
database     mysql-676f465746-97jj7                 1/1     Running   0           15m   192.168.133.193 worker-2       <none>           <none>
database     postgres-757885f655-8hv9m              1/1     Running   0           14m   192.168.133.194 worker-2       <none>           <none>
kube-system  calico-kube-controllers-5b64744bcd-65dd7 1/1     Running   0           148m   192.168.235.194 k8s-master    <none>           <none>
kube-system  calico-node-4cp69                       1/1     Running   0           142m   192.168.0.4     worker-2       <none>           <none>
kube-system  calico-node-54stq                       1/1     Running   0           140m   192.168.0.3     worker-1       <none>           <none>
kube-system  calico-node-7kfqj                       1/1     Running   0           148m   192.168.0.2     k8s-master    <none>           <none>
kube-system  calico-node-md7r5                       1/1     Running   0           143m   192.168.0.5     worker-3       <none>           <none>
kube-system  calico-node-x9hz7                       1/1     Running   0           146m   192.168.0.6     worker-4       <none>           <none>
kube-system  coredns-55cb58b774-kcsn8               1/1     Running   0           149m   192.168.235.193 k8s-master    <none>           <none>
kube-system  coredns-55cb58b774-s6tv7               1/1     Running   0           149m   192.168.235.195 k8s-master    <none>           <none>
kube-system  etcd-k8s-master                         1/1     Running   0           149m   192.168.0.2     k8s-master    <none>           <none>
kube-system  kube-apiserver-k8s-master               1/1     Running   0           149m   192.168.0.2     k8s-master    <none>           <none>
kube-system  kube-controller-manager-k8s-master      1/1     Running   0           149m   192.168.0.2     k8s-master    <none>           <none>
kube-system  kube-proxy-2nj2w                        1/1     Running   0           140m   192.168.0.3     k8s-master    <none>           <none>
kube-system  kube-proxy-c4k57                        1/1     Running   0           142m   192.168.0.4     worker-2       <none>           <none>
kube-system  kube-proxy-hnnmb                       1/1     Running   0           149m   192.168.0.2     k8s-master    <none>           <none>
kube-system  kube-proxy-pmk9g                       1/1     Running   0           143m   192.168.0.5     worker-3       <none>           <none>
kube-system  kube-proxy-wvtmq                       1/1     Running   0           146m   192.168.0.6     worker-4       <none>           <none>
kube-system  kube-scheduler-k8s-master               1/1     Running   0           149m   192.168.0.2     k8s-master    <none>           <none>
metallb-system controller-86f5578878-2ldgq               1/1     Running   1 (116m ago) 118m   192.168.38.67   worker-4       <none>           <none>
metallb-system speaker-7zvzp                      1/1     Running   0           118m   192.168.0.3     worker-1       <none>           <none>
metallb-system speaker-f9fr9                      1/1     Running   0           118m   192.168.0.2     k8s-master    <none>           <none>
metallb-system speaker-kpvpk                      1/1     Running   0           118m   192.168.0.4     worker-2       <none>           <none>
metallb-system speaker-n4766                      1/1     Running   0           118m   192.168.0.6     worker-4       <none>           <none>
metallb-system speaker-v6959                      1/1     Running   0           113m   192.168.0.5     worker-3       <none>           <none>
storage      minio-65b3b94fd6-grttz                 1/1     Running   0           9m27s   192.168.97.193 worker-3       <none>           <none>
web          apache-8875fdb8c-btxvc                  1/1     Running   0           19m   192.168.226.68 worker-1       <none>           <none>
web          apache-8875fdb8c-cfnkf                  1/1     Running   0           19m   192.168.226.67 worker-1       <none>           <none>
web          nginx-66ccff7bd8-fgdkf                  1/1     Running   0           28m   192.168.226.66 worker-1       <none>           <none>
web          nginx-66ccff7bd8-lb6dw                  1/1     Running   0           28m   192.168.226.65 worker-1       <none>           <none>
root@k8s-master:~#

```

Figure: Redis Deployment.

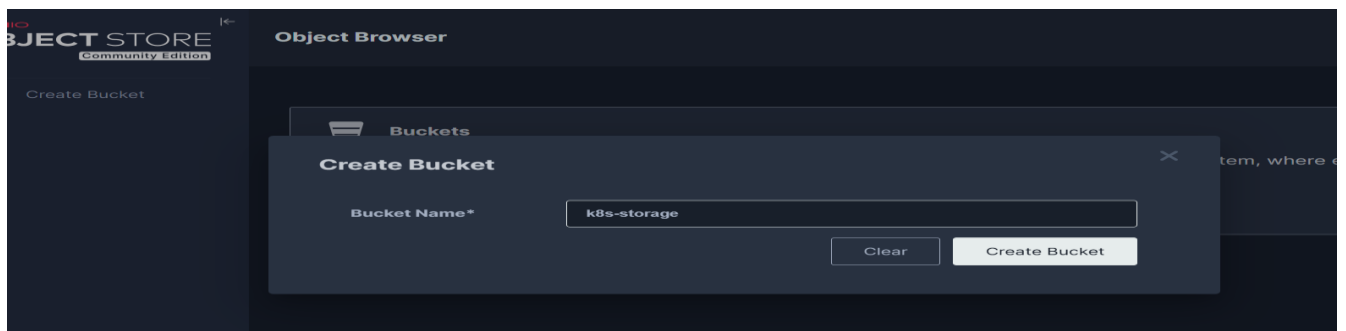
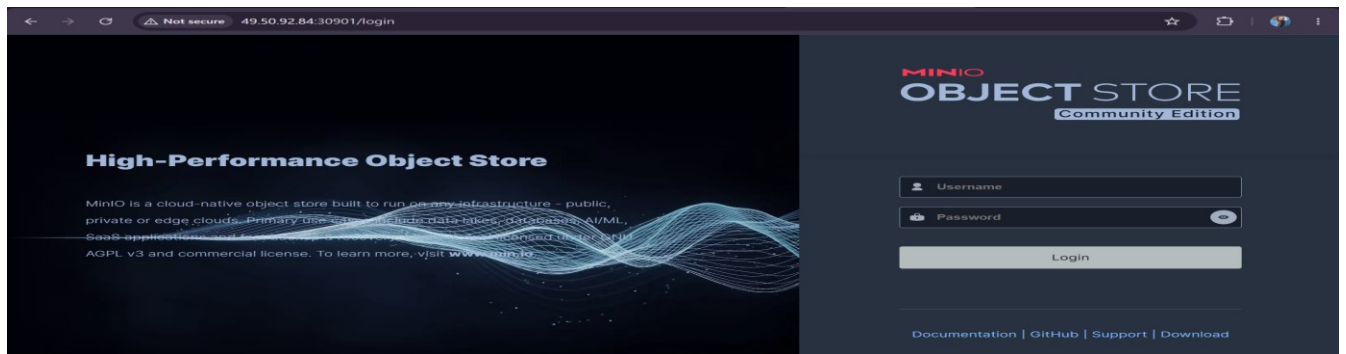


Figure: MinIO Deployment.

```

root@k8s-master:~# kubectl scale deployment nginx --replicas=5 -n web
deployment.apps/nginx scaled
root@k8s-master:~# kubectl get pods -n web -o wide
NAME                                     READY   STATUS    RESTARTS   AGE   IP              NODE           NOMINATED NODE   READINESS GATES
apache-8875fdb8c-btxvc                  1/1     Running   0           101m   192.168.226.68 worker-1       <none>           <none>
apache-8875fdb8c-cfnkf                  1/1     Running   0           101m   192.168.226.67 worker-1       <none>           <none>
nginx-66ccff7bd8-2h8tt                  1/1     Running   0           112s   192.168.226.69 worker-1       <none>           <none>
nginx-66ccff7bd8-fgdkf                  1/1     Running   0           109m   192.168.226.66 worker-1       <none>           <none>
nginx-66ccff7bd8-lb6dw                  1/1     Running   0           109m   192.168.226.65 worker-1       <none>           <none>
nginx-66ccff7bd8-nnrxh                  1/1     Running   0           112s   192.168.226.71 worker-1       <none>           <none>
nginx-66ccff7bd8-tff292                 1/1     Running   0           112s   192.168.226.70 worker-1       <none>           <none>
root@k8s-master:~#

```

Figure: Scaling

```

root@k8s-master:~# kubectl delete pod nginx-66ccff7bd8-fgdkt -n web
pod "nginx-66ccff7bd8-fgdkt" deleted
root@k8s-master:~# kubectl get pods -n web -w
NAME                                READY    STATUS    RESTARTS   AGE
apache-8875fdb8c-btxvc              1/1      Running   0           103m
apache-8875fdb8c-cfnkf              1/1      Running   0           103m
nginx-66ccff7bd8-2h8tt              1/1      Running   0           3m54s
nginx-66ccff7bd8-lb6dw              1/1      Running   0           111m
nginx-66ccff7bd8-nnrxx              1/1      Running   0           3m54s
nginx-66ccff7bd8-qdpjl              1/1      Running   0           9s
nginx-66ccff7bd8-tf292              1/1      Running   0           3m54s

```

Figure: Self-Healing

```

root@k8s-master:~# kubectl rollout undo deployment nginx -n web
deployment.apps/nginx rolled back
root@k8s-master:~#

```

```

waiting for deployment "nginx" rollout to finish: 3 out of 3 new replicas have
^Croot@k8s-master:~# kubectl scale deployment nginx --replicas=2 -n web
deployment.apps/nginx scaled
root@k8s-master:~# kubectl rollout undo deployment nginx -n web
deployment.apps/nginx rolled back
root@k8s-master:~# kubectl rollout status deployment nginx -n web
deployment "nginx" successfully rolled out
root@k8s-master:~#

```

Figure: Rolling Updates

```

root@k8s-master:~# kubectl rollout undo deployment nginx -n web
deployment.apps/nginx rolled back
root@k8s-master:~#

```

Figure: Rollback

```

root@k8s-master:~# nano mysql-pv.yaml
root@k8s-master:~# kubectl apply -f mysql-pv.yaml
persistentvolume/mysql-pv created
root@k8s-master:~# kubectl get pv
NAME      CAPACITY  ACCESS MODES  RECLAIM POLICY  STATUS    CLAIM  STORAGECLASS  VOLUMEATTRIBUTESCLASS
mysql-pv  5Gi       RWO           Retain          Available    


```

```

root@k8s-master:~# nano mysql-pvc.yaml
root@k8s-master:~# kubectl apply -f mysql-pvc.yaml
persistentvolumeclaim/mysql-pvc created
root@k8s-master:~# kubectl get pvc -n database
NAME      STATUS  VOLUME  CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS
mysql-pvc Bound    mysql-pv  5Gi       RWO             


```

```

root@k8s-master:~# kubectl get pvc -n database
NAME      STATUS  VOLUME  CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS  AGE
mysql-pvc Bound    mysql-pv  5Gi       RWO             

root@k8s-master:~#

```

Figure: Persistent storage



Figure: Website Deployment.

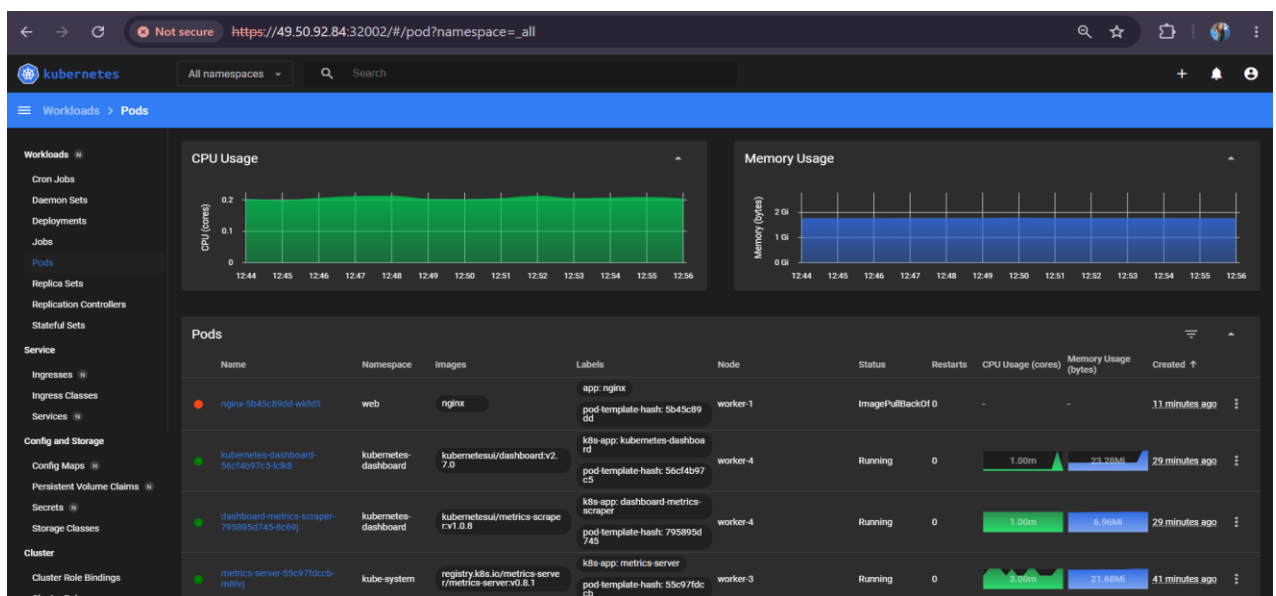
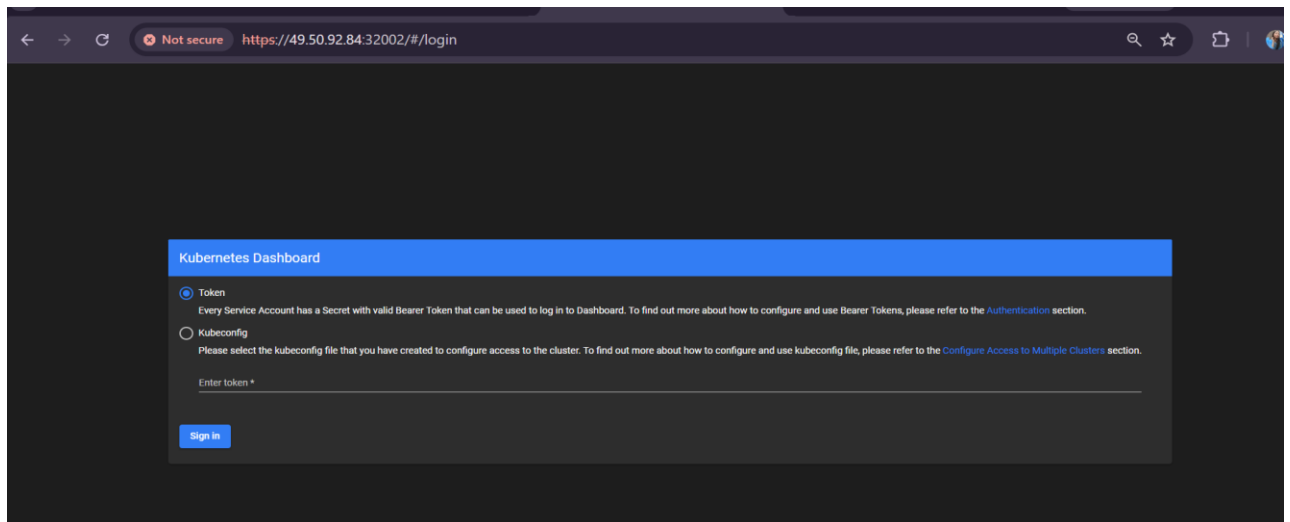
```
root@k8s-master:~# kubectl top nodes
```

NAME	CPU (cores)	CPU%	MEMORY (bytes)	MEMORY%
k8s-master	71m	1%	3103Mi	31%
worker-1	40m	1%	2366Mi	24%
worker-2	73m	1%	3089Mi	31%
worker-3	35m	0%	2280Mi	23%
worker-4	53m	1%	2326Mi	23%

```
root@k8s-master:~# kubectl top pods -A
```

NAMESPACE	NAME	CPU (cores)	MEMORY (bytes)
cache	redis-7b8446847d-hhtx5	3m	5Mi
database	mysql-7f5569f55d-gnlmn	5m	390Mi
database	postgres-757885f655-8hv9m	1m	31Mi
kube-system	calico-kube-controllers-5b64744bcd-65dd7	4m	27Mi
kube-system	calico-node-4cp69	29m	135Mi
kube-system	calico-node-54stq	32m	135Mi
kube-system	calico-node-7kfjq	17m	136Mi
kube-system	calico-node-md7r5	17m	135Mi
kube-system	calico-node-x9hz7	31m	136Mi
kube-system	coredns-55cb58b774-kcsn8	1m	14Mi
kube-system	coredns-55cb58b774-s6tv7	1m	14Mi
kube-system	etcd-k8s-master	15m	58Mi
kube-system	kube-apiserver-k8s-master	35m	293Mi
kube-system	kube-controller-manager-k8s-master	8m	56Mi
kube-system	kube-proxy-2nj2w	1m	13Mi
kube-system	kube-proxy-c4k57	1m	12Mi
kube-system	kube-proxy-hnnmb	1m	13Mi
kube-system	kube-proxy-pmk9g	1m	12Mi
kube-system	kube-proxy-wvtmq	1m	12Mi
kube-system	kube-scheduler-k8s-master	2m	19Mi
kube-system	metrics-server-55c97fdccb-m8hrj	2m	22Mi
kubernetes-dashboard	dashboard-metrics-scraper-795895d745-8c69j	1m	8Mi
kubernetes-dashboard	kubernetes-dashboard-56cf4b97c5-lc1k8	1m	11Mi
storage	minio-694b84fdb8-gcbfj	1m	102Mi
web	apache-8875fdb8-btxvc	1m	7Mi
web	apache-8875fdb8-cfnkf	1m	8Mi
web	nginx-5bcbf5f8d6-7wff4	0m	4Mi
web	nginx-5bcbf5f8d6-c4s79	0m	4Mi

Figure: Metrics server



The screenshot shows the "Persistent Volume Claims" page in the Kubernetes Dashboard. The browser address bar displays "https://49.50.92.84:32002/#/persistentvolumeclaims". The page has a dark theme. The left sidebar shows the navigation menu with "Persistent Volume Claims" selected. The main content area has a blue header bar with the text "Config And Storage > Persistent Volume Claims". Below this, there is a table of Persistent Volume Claims.

Name	Namespace	Labels	Status	Volume	Capacity	Access Modes	Storage Class	Created ↑
mysql-pvc	database	-	Bound	mysql-pv	5Gi	ReadWriteOnce	-	2 days ago

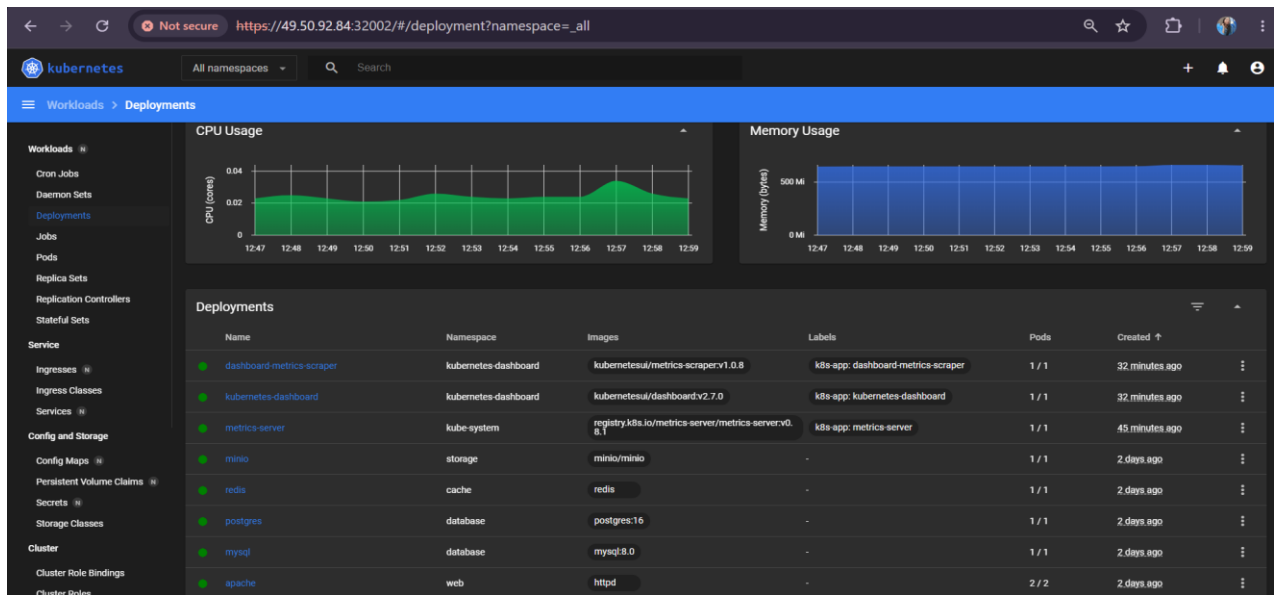


Figure: Kubernetes dashboard